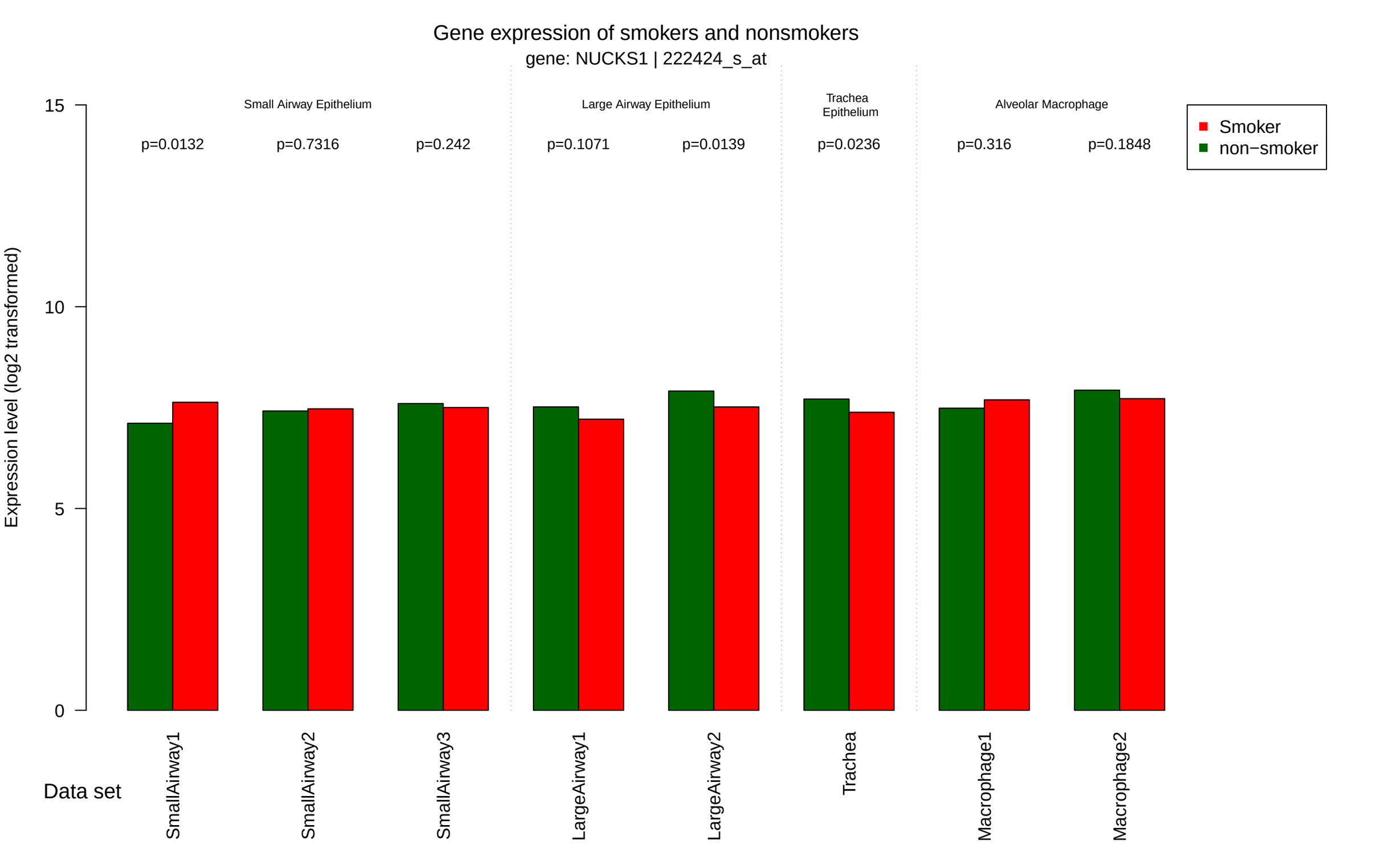
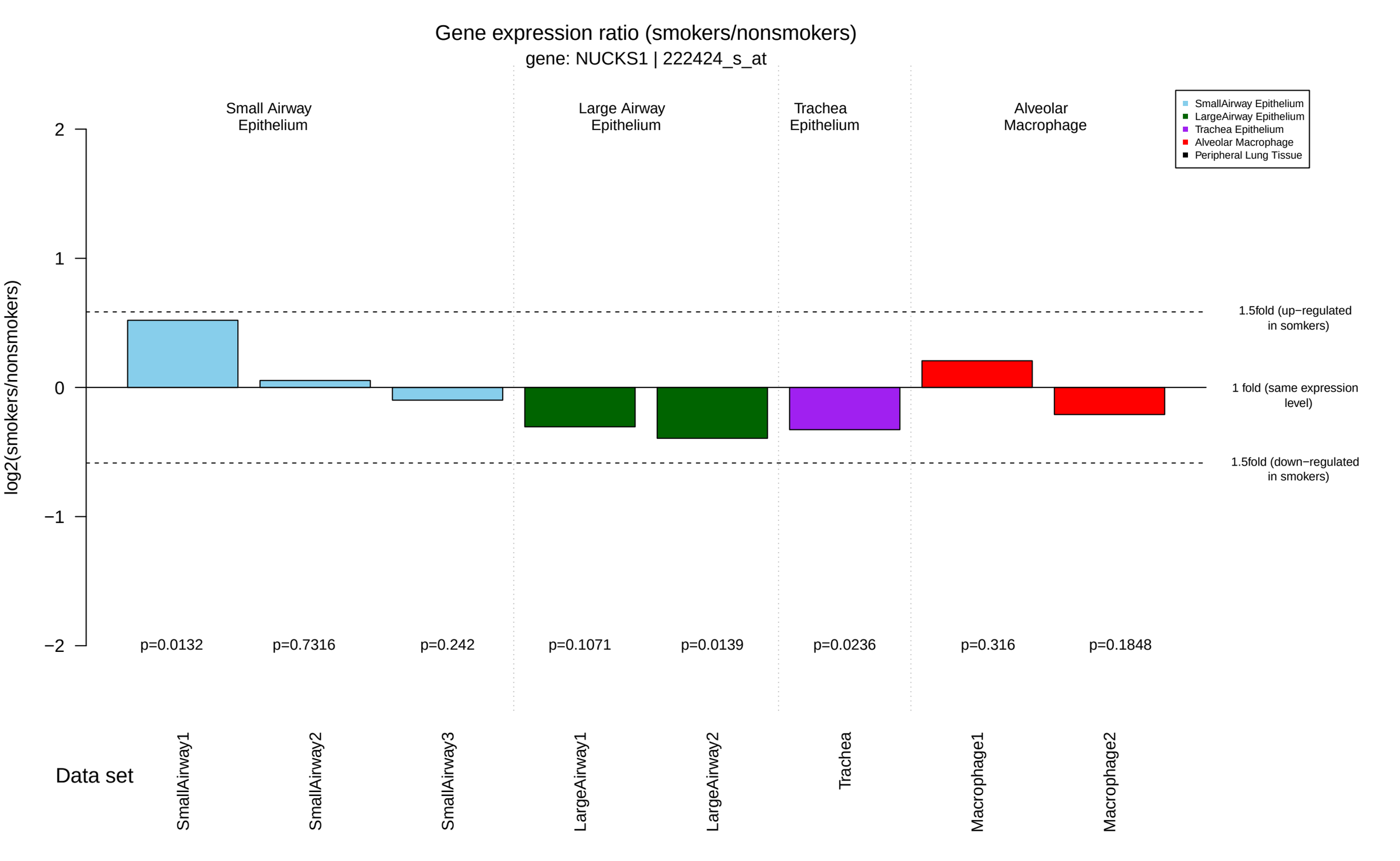
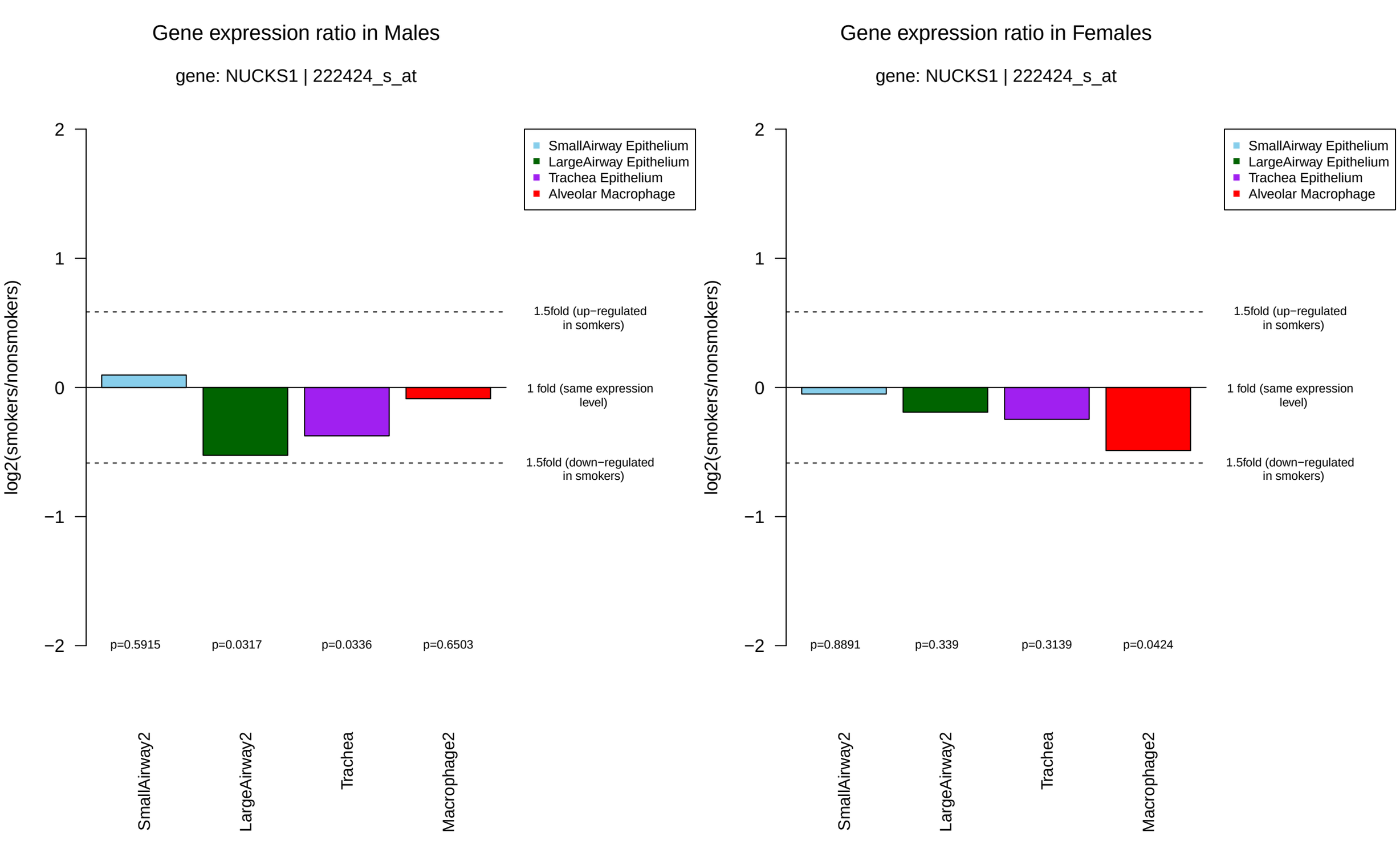
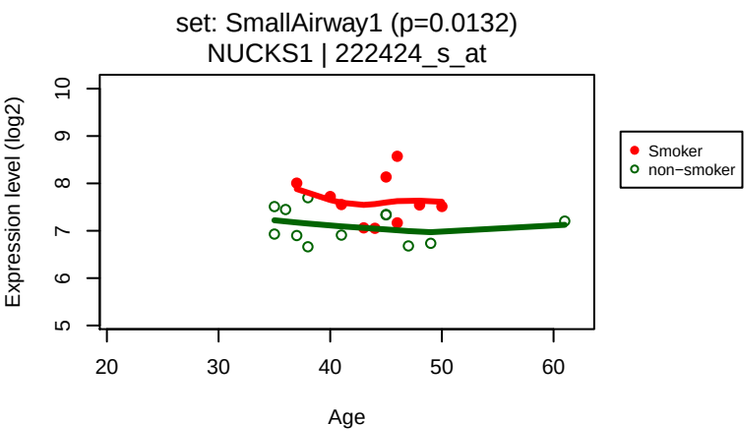
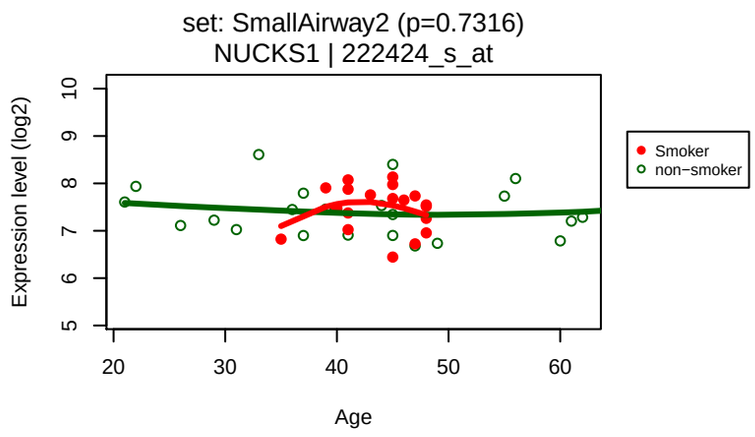
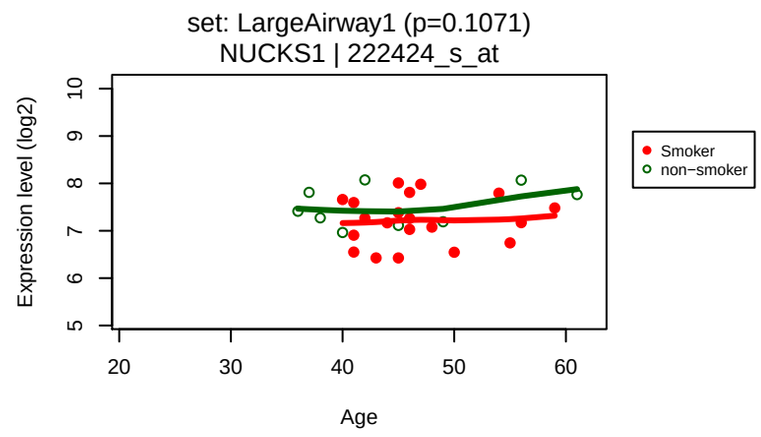

Gene expression of smokers and nonsmokers
gene: NUCKS1 | 222424_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0132
p=0.7316
p=0.242
p=0.1071
p=0.0139
p=0.0236
p=0.316
p=0.1848
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

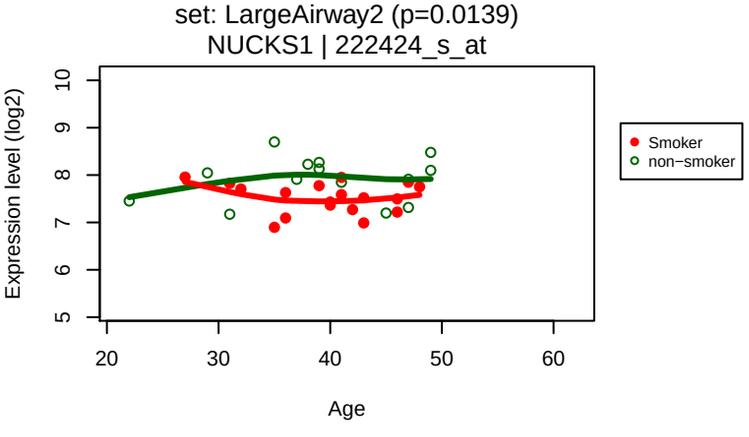

Gene expression ratio (smokers/nonsmokers)
gene: NUCKS1 | 222424_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0132
p=0.7316
p=0.242
p=0.1071
p=0.0139
p=0.0236
p=0.316
p=0.1848
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

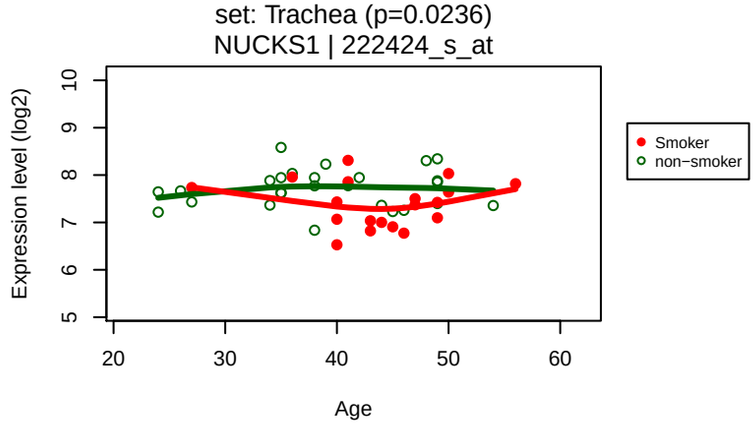

Gene expression ratio in Males
gene: NUCKS1 | 222424_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5915
p=0.0317
p=0.0336
p=0.6503
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NUCKS1 | 222424_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8891
p=0.339
p=0.3139
p=0.0424
SmallAirway2
LargeAirway2
Trachea
Macrophage2

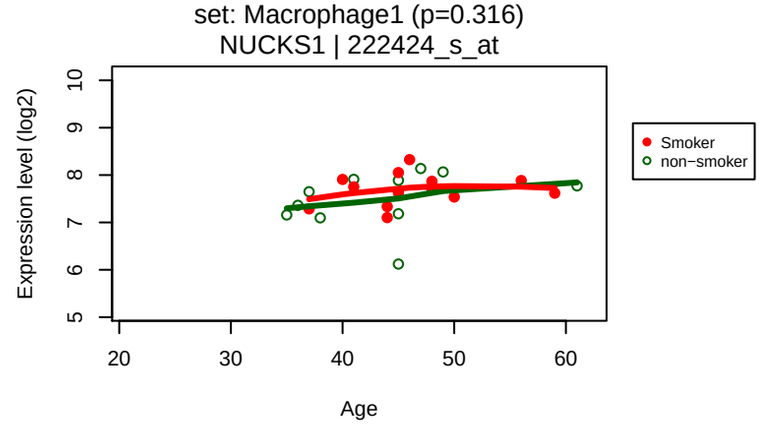

set: SmallAirway1 (p=0.0132)
NUCKS1 | 222424_s_at
Expression level (log2)
Age
Smoker
non−smoker

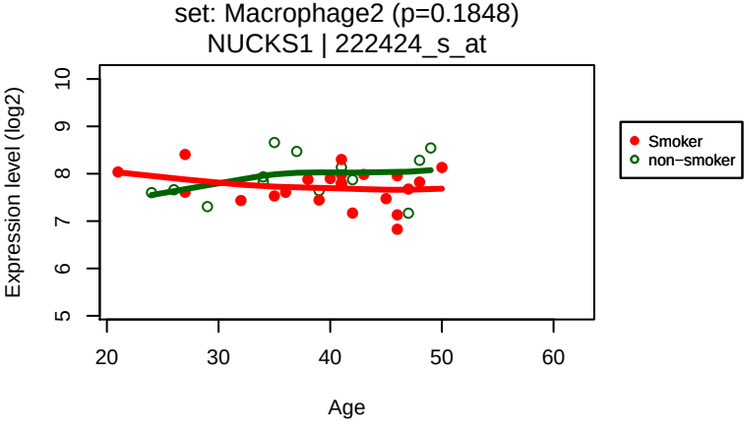

set: SmallAirway2 (p=0.7316)
NUCKS1 | 222424_s_at
Expression level (log2)
Age
Smoker
non−smoker

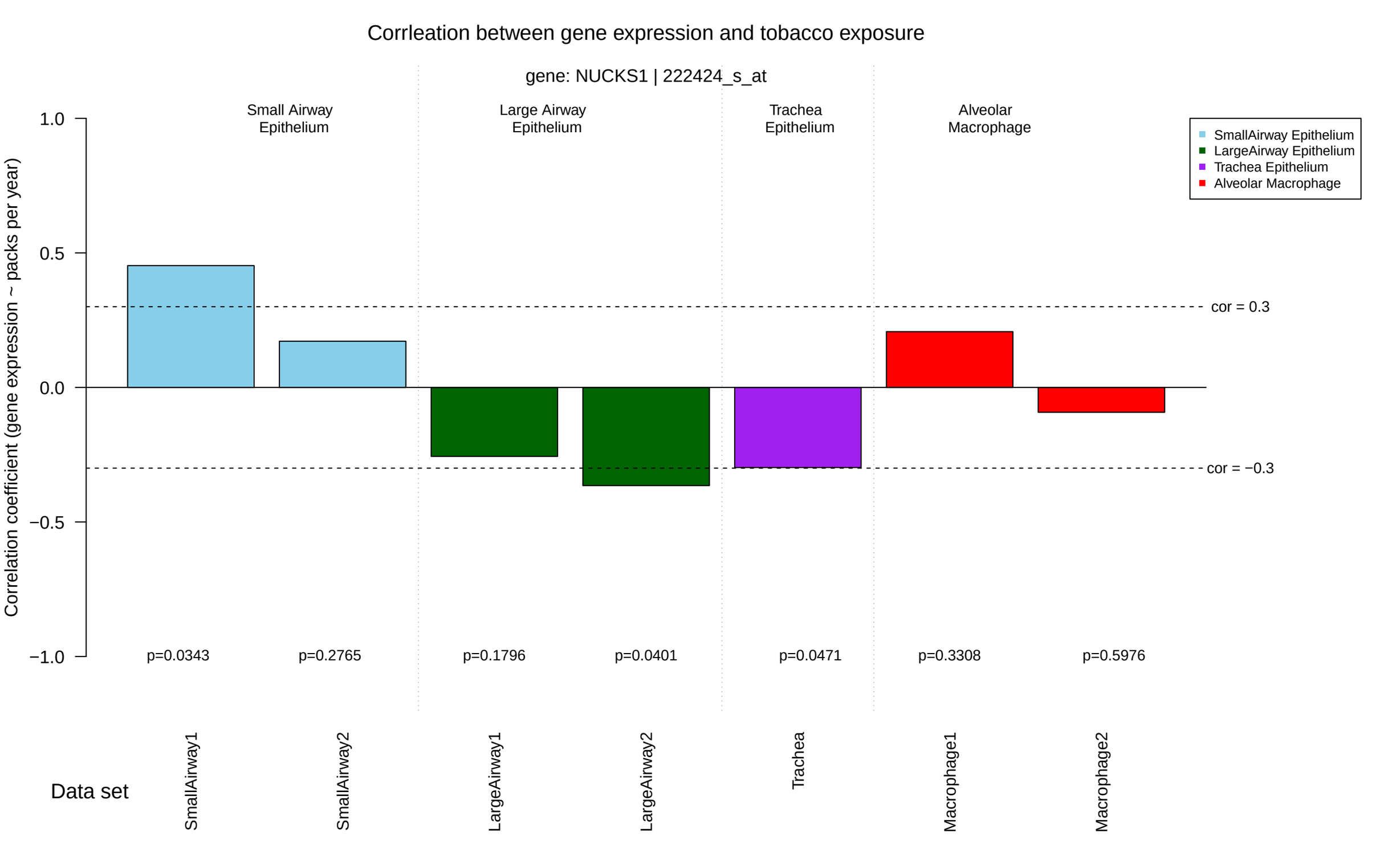

set: LargeAirway1 (p=0.1071)
NUCKS1 | 222424_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0139)
NUCKS1 | 222424_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0236)
NUCKS1 | 222424_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.316)
NUCKS1 | 222424_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1848)
NUCKS1 | 222424_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NUCKS1 | 222424_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0343
p=0.2765
p=0.1796
p=0.0401
p=0.0471
p=0.3308
p=0.5976
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2